Automated Quality Inspection System using Computer Vision

Abstract

This project uses OpenCV and Python to automatically inspect manufactured parts and identify defects using image processing techniques. The system performs real-time analysis through a camera feed and classifies products as PASS or FAIL.

Objective

- Automate quality inspection.
- Reduce manual inspection errors.
- Improve manufacturing efficiency.

Problem Statement

Manual inspection is slow and error-prone. Industries require automated systems capable of detecting defects accurately and consistently.

Software Used

- Python
- OpenCV
- NumPy

Hardware Used

- Webcam
- Computer/Laptop

Methodology

1. Capture live video feed.
2. Convert image to grayscale.
3. Apply Gaussian Blur.
4. Apply thresholding.
5. Detect contours.
6. Identify defect regions.
7. Generate PASS/FAIL result.

Working Principle

The camera continuously captures images. OpenCV processes each frame to identify abnormal regions. If a defect exceeds a predefined threshold area, the system labels the object as defective.

Flow of System

Camera → Image Processing → Defect Detection → PASS/FAIL Decision

Results

The system successfully detected visible defects and highlighted them using bounding boxes.

Advantages

- Fast inspection
- Low cost
- Real-time monitoring
- Reduced human error

Future Scope

- AI-based defect detection
- YOLO object detection
- Industrial conveyor belt integration
- Automatic sorting system

Conclusion

The project demonstrates how computer vision can automate quality control and improve manufacturing productivity.

References

- OpenCV Documentation
- Python Documentation
- Industrial Inspection Research Papers